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Precision agriculture in the age of AI: A systematic review of machine learning methods for crop disease detection

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ABSTRACT

Artificial Intelligence (AI) has become a critical tool in modern precision agriculture, particularly in the detection of plant diseases and pests. This study provides a comprehensive review of current AI methodologies applied to crop disease detection, with a focus on machine learning models, dataset availability, and performance metrics. Our findings indicate that Convolutional Neural Networks (CNNs) are the most widely used and cost-effective approach, while Vision Transformers (ViTs) exhibit superior accuracy but require significantly higher computational resources. We identify key research gaps, including the geographic bias in dataset origins, the trade-off between data quality and quantity, and the limited exploration of hybrid AI models. Additionally, challenges related to the scalability and real-time applicability of AI solutions in resource-constrained agricultural environments highlight the need for more efficient computational techniques. Cost-effectiveness remains a concern, as high-performance models often demand expensive infrastructure and expertise, limiting their adoption by smallholder farmers. Furthermore, cybersecurity risks in digital agriculture and disparities in research focus across different crops and disease types underscore the necessity for broader investigations. Unlike prior reviews, this study uniquely integrates methodological performance, dataset characteristics, and cybersecurity perspectives, offering a holistic synthesis that highlights not only technical advances but also practical barriers to large-scale adoption. Future work should prioritize model robustness, efficiency, and accessibility.

Introduction

The global population is projected to exceed nine billion by mid-century, causing a substantial rise in the demand for food and placing ever-greater pressures on the agricultural sector to boost output [1,2]. Meeting this demand is complicated by a wide range of concerns, including labor shortages, which are partly driven by rural-to-urban migration [3,4]. At the same time, environmental issues, such as global warming, biodiversity loss, and soil degradation, demand that food production become more sustainable [5]. Seasonal variability [6] and extreme climate events, from droughts to floods, further exacerbate the volatility of growing conditions and heighten the likelihood of crop failures [7]. Moreover, rising temperatures and altered precipitation regimes are accelerating the emergence and geographic expansion of crop pathogens, while biodiversity loss reduces the natural biological barriers that suppress disease outbreaks, thereby increasing disease pressure in the field [8,9]. These converging factors underscore the need for rapid, accurate, and scalable disease-detection methods to enable timely interventions and protect yields in increasingly unpredictable agro-ecosystems.

Pests and diseases pose one of the most pressing challenges. Collectively, they account for nearly 30 % of global crop losses, underscoring the urgency of developing effective and timely disease detection and management strategies [10]. Traditional scouting and identifying diseases rely heavily on human expertise, which is time-consuming, expensive, and often imprecise [11]. In low-income regions, limited access to agricultural experts can lead to further yield loss and raise the specter of food insecurity.

In response to these threats, precision agriculture has gained traction as a data-driven framework for optimizing farming inputs, such as fertilizers, pesticides, and water, to specific field conditions. By leveraging sensors, robotics, and intelligent computing, precision agriculture aims

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to maximize efficiency and sustainability while minimizing environmental impact [12]. Tools like drones and satellites equipped with remote sensing technologies can capture real-time information on abiotic and biotic stresses [13]. In contrast, multispectral or hyperspectral imaging provides even more nuanced insights into plant health [14].

A key component of this high-tech transformation is artificial intelligence (AI), particularly when combined with computer vision. AI-driven image analysis enables the automated detection of early-stage disease symptoms, such as leaf discoloration or fungal spots, at a speed and scale that far outperforms human observation [15]. This approach can also help reduce pesticide usage by allowing targeted, rather than blanket, chemical applications—mitigating the risks posed to both human health and the environment [16]. Moreover, improved computational power and ever-larger datasets continue to strengthen machine learning and deep learning models, enabling more accurate diagnosis of plant pathologies. However, data quality remains a pivotal factor in model performance; inaccurate or non-representative datasets can lead to poor generalizability.

Alongside these technical and environmental challenges, emerging cybersecurity risks threaten digital agriculture systems. Hacking, data manipulation, and other forms of cyber warfare can disrupt farm operations or undermine the trustworthiness of AI-driven recommendations [17]. In AI-based crop-disease detection pipelines, adversarial image attacks or model-poisoning could generate false diagnoses or conceal early outbreaks, delaying intervention and amplifying economic losses [18,19]. Because these pipelines rely on networked field sensors, drones, and cloud services, every communication hop presents a potential attack surface. A single compromised imaging sensor can feed corrupted data into the training stream, gradually shifting model weights until healthy leaves appear diseased, or vice-versa. Similarly, ransomware aimed at agricultural IoT devices could block real-time scouting data, forcing farmers to make decisions blind. Ensuring robust data protection, secure communication channels, and continuous model-integrity monitoring is, therefore, an important dimension of modern precision agriculture [20].

Given the converging need for greater productivity, reduced chemical inputs, and stronger resilience against climate volatility, a systematic review of AI-based crop disease detection methods is both timely and necessary. By synthesizing current advancements in AI-driven techniques and examining practical case studies, researchers and practitioners can better understand where these technologies excel and where gaps still exist, especially in low-resource settings or regions prone to severe pest outbreaks.

In the following sections, a concise definition of precision agriculture and its technological underpinnings is offered, illustrating how site-specific data collection and analysis can be used to drive informed decision-making in diverse farming contexts. Next, an overview of commonly employed machine learning (ML) and computer vision (CV) methods for plant disease detection is provided, followed by a balanced examination of their primary strengths, potential limitations, and emerging security considerations.

We then detail the methodology used in our systematic literature review, clarifying our search strategy, inclusion and exclusion criteria, data extraction processes, and the quality assessment techniques employed to ensure robust and unbiased findings. In the results section, we present our key findings, demonstrating how ML and CV-based disease detection models have evolved in terms of algorithmic sophistication, computational requirements, and real-world deployment scenarios. We highlight critical performance metrics (e.g., accuracy, precision, recall, F1-score) reported in the literature, along with insights into model scalability, cost-effectiveness, and environmental impact. Following that, our discussion section interprets the implications of these findings in the broader context of precision agriculture. We compare different AI-driven approaches, emphasize best practices that emerged from the studies, and identify knowledge gaps where future work could yield advances. Finally, we conclude by summarizing the key contributions of this review, reaffirming the potential of AI-driven disease detection to address critical agricultural challenges, and proposing directions for further research.

Contributions

This review provides a systematic and comprehensive synthesis of 153 original research articles published between 2016 and 2024, offering an up-to-date overview of methodological advancements across classical machine learning, convolutional neural networks, vision transformers, and hybrid approaches. Our work compares performance metrics such as accuracy, precision, recall, and F1-score but also places emphasis on cost-effectiveness, scalability, and real-world applicability in resource-constrained agricultural settings. We also examined the role of dataset characteristics, including origin, size, quality, and annotation standards, in shaping model performance and generalizability. We highlight the geographic imbalance of available datasets and the overrepresentation of certain staple crops, underscoring critical research gaps that limit the broader applicability of AI models. An emerging but underexplored dimension in this domain, the cybersecurity risks associated with digital agriculture systems, including adversarial attacks, data manipulation, and model poisoning was also investigated. By integrating methodological, practical, and security perspectives, this review offers a holistic framework that informs future research and provides actionable insights for developing robust, accessible, and secure AI solutions in precision agriculture.

Background

Precision agriculture is an innovative approach to farming that leverages advanced technologies to optimize agricultural practices, improve resource efficiency, and enhance crop yields. This methodology is centered around tailoring farming interventions to the specific conditions of individual fields or sections of fields, enabling sustainable and highly efficient agricultural management. By employing tools such as remote sensing, geographic information systems (GIS), and machine learning, precision agriculture addresses the growing global demand for food while minimizing environmental impact [21].

Technological methods are central to precision agriculture, facilitating data collection and analysis to inform decision-making processes. Remote sensing technologies, including satellites, drones, and hyperspectral cameras, play a vital role in capturing detailed images of crops to monitor their health and detect potential diseases. For instance, hyperspectral imaging has been employed to assess disease severity in crops, enabling farmers to address issues at an early stage [22]. Additionally, the integration of artificial intelligence, particularly deep learning algorithms, has revolutionized crop disease detection by automatically identifying patterns and anomalies in images. These algorithms, such as convolutional neural networks (CNNs), are trained on large datasets to enhance their accuracy and adaptability to diverse agricultural scenarios [23].

A critical aspect of precision agriculture is site-specific data collection, which ensures that interventions are tailored to the unique conditions of each area within a field. Data is gathered using IoT sensors, UAVs (unmanned aerial vehicles), and other tools that monitor soil moisture, temperature, and nutrient levels in real time. For example, UAVs equipped with multispectral cameras are frequently used to identify stressed zones within crops that require immediate attention [24]. These tools allow for the detailed mapping of field variability, essential for optimizing inputs such as fertilizers and pesticides.

The vast amount of data collected through these technologies necessitates advanced data analysis techniques to transform raw information into actionable insights. Machine learning and big data analytics are employed in precision agriculture to process and interpret complex datasets. These methods enable predictions regarding disease outbreaks, yield estimation, and resource optimization. For instance, predictive

models based on machine learning have been used to forecast crop health and assess the impact of environmental factors, providing farmers with valuable tools for proactive management [25]. These analytical tools allow for a comprehensive understanding of crop performance, ensuring that resources are allocated efficiently.

Aim

This systematic review aims to comprehensively analyze the application of artificial intelligence (AI), particularly machine learning and deep learning techniques, in detecting and managing major crop diseases. The review seeks to identify which AI-driven methods have demonstrated the most promising performance across key metrics such as accuracy, precision, scalability, and cost-effectiveness. By comparing and contrasting the outcomes of different models, this study critically evaluates their practical utility in real-world agricultural settings, especially in low-resource environments. It further considers the trade-offs between high-performance algorithms like Vision Transformers and the more scalable, accessible architectures such as convolutional neural networks.

Additionally, the review explores how dataset-related characteristics, such as origin, size, image quality, and annotation standards, influence the reliability and generalizability of AI-based disease detection systems. By mapping the distribution of studied crops and trees, the review also sheds light on research disparities, highlighting the dominance of certain economically important species like rice, tomato, and potato, while underscoring the need for broader crop inclusion and regional representation. Ultimately, this work synthesizes current technological advances and outlines future directions for more inclusive, secure, and cost-effective AI solutions in precision agriculture. To address these intertwined challenges, the present study poses the following research questions:

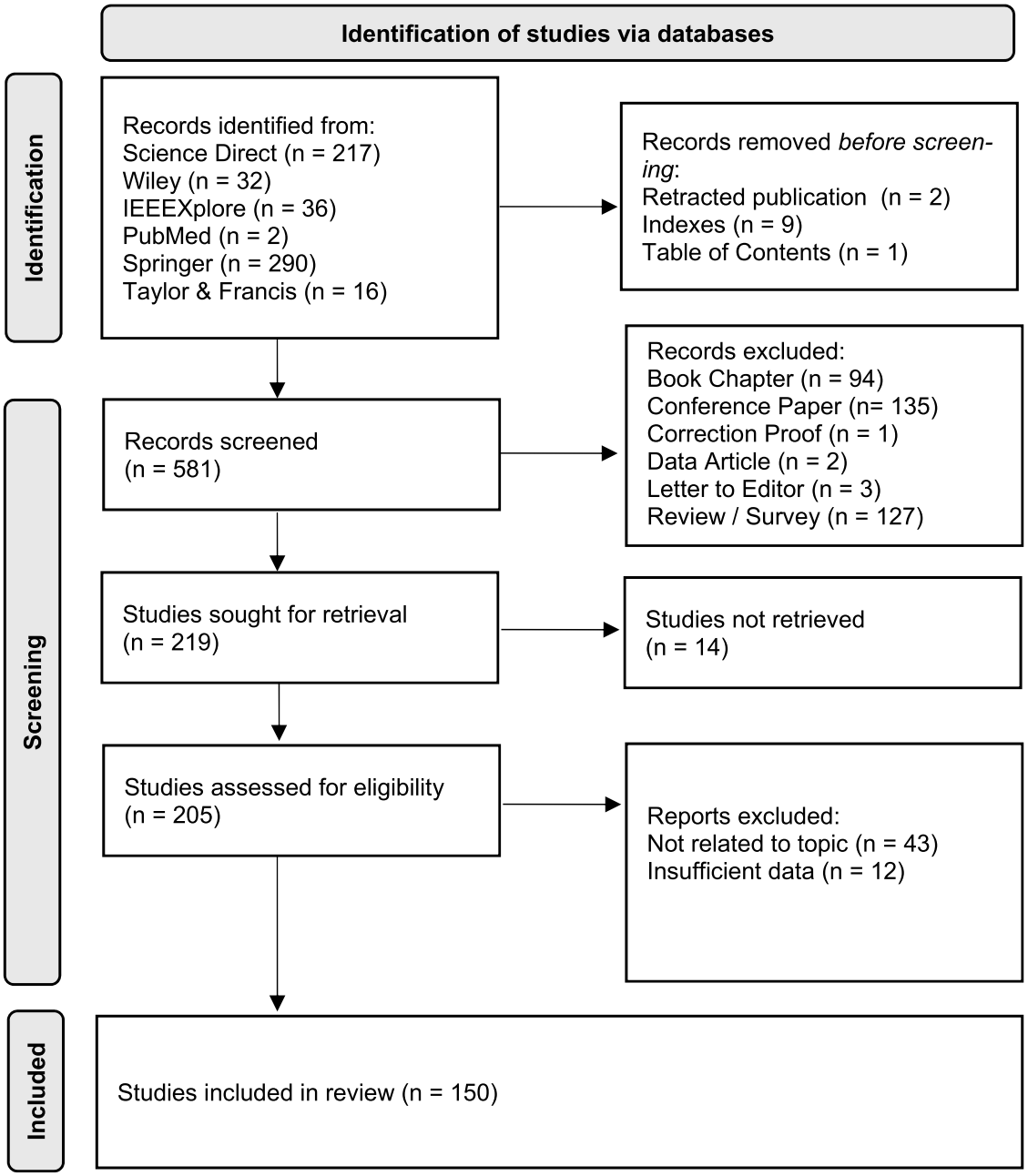


Fig. 1. Identification of studies to include in the analysis.

1. Which AI-driven techniques (including machine learning and deep learning) have been most effective in detecting and managing major crop diseases, and how do they compare regarding accuracy, precision, scalability, and cost-effectiveness?
2. How do data-related factors—such as dataset origin, annotation standards, and size—impact the performance and reliability of AI-based disease detection systems?
3. Which types of trees and crops have been most extensively studied in disease detection using artificial intelligence and image analysis, and what factors explain the disparity in research volume among them?

Methods

Framework

This systematic review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework, which provides standardized guidelines to ensure transparency, reproducibility, and methodological rigor in literature reviews. By adhering to the PRISMA protocol, we structured the review process into four key stages: identification, screening, eligibility, and inclusion. This framework guided the development of our search strategy, the application of inclusion and exclusion criteria, and the systematic extraction of data. To complement the textual description, a PRISMA-style flow diagram (Fig. 1) has been included to provide a graphical overview of the study selection process.

Search strategy

A comprehensive search was conducted to identify studies focusing on the application of artificial intelligence (AI), machine learning (ML), and deep learning (DL) in plant and crop disease detection within the realm of precision agriculture.

The literature search was systematically conducted across six major academic databases: (1) ScienceDirect, (2) Wiley Online Library, (3) IEEE Xplore, (4) PubMed, (5) SpringerLink, and (6) Taylor & Francis Online. These selection ensure broad coverage of peer-reviewed research in computer science, engineering, and agricultural sciences. These databases were chosen because they collectively index the vast majority of high-quality journals, thereby providing a comprehensive and reliable evidence base for the review.

The following search query was employed:

("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning") AND ("plant disease detection" OR "crop disease detection") AND ("image processing" OR "multispectral imaging") AND ("precision agriculture" OR "smart farming")

Inclusion and exclusion criteria

Studies were selected based on the following inclusion criteria:

- Focus on AI, ML, or DL applications in plant or crop disease detection.
- Utilization of image processing or multispectral imaging techniques.
- Relevance to precision agriculture or smart farming practices.
- Peer-reviewed.
- Original research
- Published in one of the selected databases.
- Written in English.

Exclusion criteria encompassed:

- Non-peer-reviewed articles, editorials, and opinion pieces.
- Studies lack empirical data or methodological rigor.
- Retracted publications and index listings.
- Other publishers.
- Language other than English.
- Not research articles (reviews, book chapters, letter to editor, shorts).
- Not full research (conference papers).

Study selection

The initial search across six academic databases yielded a total of 593 records: Science Direct ($n = 217$), Springer ($n = 290$), IEEEXplore ($n = 36$), Wiley ($n = 32$), Taylor & Francis ($n = 16$), and PubMed ($n = 2$). Before the screening phase, 12 records were removed, including 2 retracted publications, 9 index-only entries, and 1 table of contents item, resulting in 581 records eligible for initial screening. These records were then evaluated based on title and abstract to determine their relevance to the predefined inclusion criteria. During this phase, 362 records were excluded: 94 were book chapters, 135 were conference papers, 1 was a correction proof, 2 were data articles, 3 were letters to the editor, and 127 were review or survey papers, as the focus of this review was restricted to original research articles presenting primary data.

Following the screening, 219 full-text articles were sought for retrieval. Of these, 14 could not be retrieved due to access limitations or broken links, leaving 205 articles for full-text assessment. After applying the eligibility criteria at this stage, an additional 55 articles were excluded, 43 for not being directly related to the research topic and 12 due to insufficient methodological or performance data. Ultimately, 150 studies met all inclusion criteria and were included in the final synthesis of this systematic review. Discrepancies throughout the selection process were resolved by consensus among the reviewers, ensuring consistency and rigor in the study selection.

Although an extensive number of studies met every inclusion criterion, the number 150 is not a pre-set limit; instead, it reflects the complete body of eligible, peer-reviewed primary research available across the six databases at the time of search (January – March 2025). Including the full set of 150 eligible studies maximises statistical power for subgroup analyses comparing classical machine-learning algorithms, CNNs, Vision Transformers, and hybrid approaches across diverse crops and imaging modalities. The sample spans nine consecutive publication years (2016–2024) and 26 countries, providing a broad geographical and temporal perspective that enhances external validity. The number remains feasible for data extraction and critical appraisal by the review team, avoiding the selection bias that could arise from down-sampling.

Data extraction

The extracted data included the study objectives and hypotheses and the technological methodologies employed (machine learning or deep learning algorithms used in each study). Information on data collection techniques was gathered, including methods for image acquisition such as satellite imaging, UAVs, or multispectral sensors. Data-related parameters were also extracted, such as the number of images used, type of image, and quality or image size. Data analysis procedures were recorded, along with the software tools utilized to process and analyze the data and how data annotation was done. The types of trees and crops were also recorded, as well as the type of disease or sub-optimal status that the study aimed to identify.

Key findings were extracted from each study, including performance metrics such as accuracy, precision, and F1-score, as well as scalability and cost-effectiveness, which provided insights into the effectiveness of the applied methods in crop disease detection. These metrics were used to evaluate and compare the overall performance of the technological approaches across different studies. The extracted data were organized into a structured table to facilitate statistical analysis and reporting. The included studies can be found in Table 1 as supplement.

Results

Our search resulted in 593 articles, which were further reduced to 150 after the exclusion criteria were applied. Although our research

query did not impose any time constraints, the earliest included publications date back to 2016, suggesting that the application of AI in crop disease detection is a relatively recent yet rapidly evolving field. The number of eligible studies rose from a single article in 2016 to 64 in 2024, reflecting rapid adoption of AI-based crop-disease detection. Data is shown in Fig. 2.

In the next paragraphs, we will answer our research questions based on the obtained data.

1) Which AI-driven techniques (including machine learning and deep learning) have been most effective in detecting and managing major crop diseases, and how do they compare regarding accuracy, precision, scalability, and cost-effectiveness?

The number of articles included allowed the analysis of several AI models for disease detection in crops. Classical machine learning models, such as SVM or KNN, were applied in 19 studies [26–44]. Deep learning models based on Convolutional Neural Networks (CNNs) were utilized in 49 studies [45–90]. Additionally, the most recent Vision Transformers (ViTs) were explored in 9 studies [61,91–96]. Furthermore, some authors opted for hybrid AI approaches, integrating CNNs with classical ML models such as SVM, which was observed in 2 studies [97,98]. This diversity in AI model selection highlights the evolution of machine learning in plant disease detection, where deep learning models, especially CNNs and ViTs, are increasingly favored due to their superior feature extraction capabilities.

The preference for CNNs, particularly custom architectures, suggests that researchers are optimizing feature extraction for domain-specific agricultural applications. Additionally, transfer learning remains a relevant approach for leveraging pre-trained models while reducing computational requirements. The integration of classical ML models with deep learning demonstrates efforts to improve generalization, optimize performance, and enhance model interpretability, particularly when working with small datasets or structured agricultural data.

The results of our analysis confirm that deep learning models generally outperform classical machine learning models in terms of accuracy and precision. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) consistently achieved high classification performance, with several studies reporting accuracies above 99 % [75,89,94,95]. In particular, ViT-based models demonstrated remarkable accuracy, with some studies achieving values close to 100 % [106–108]. CNN-based approaches also demonstrated strong performance, often exceeding 99 %, especially when optimized through transfer learning [109–112]. Fig. 3 shows mean accuracy across these three main AI architectures. In contrast, classical machine learning models such as SVM and KNN, while still effective, typically showed lower accuracy levels compared to deep learning approaches. This gap stems from their reliance on hand-crafted feature descriptors such as color histograms and texture metrics that capture less of the fine-grained, multi-scale patterns characteristic of plant-disease imagery. Moreover, most classical models were trained from scratch on relatively small, sometimes imbalanced datasets, whereas CNN- and ViT-based studies frequently leveraged large-scale pre-training and extensive data-augmentation pipelines, enhancing generalizability across variations in illumination, background, and crop species. These findings reinforce the growing trend toward deep learning architectures in plant disease detection, highlighting their superior feature extraction capabilities and adaptability to complex datasets.

Not all studies report information on cost-effectiveness and scalability. However, among the studies where this data is available [28,29,74-76,94,98-102], it is evident that a significant proportion utilize RGB images from publicly available datasets [28,80-82,103-106], which reduces costs associated with data collection. Additionally, most cost-effective models rely on CNNs trained through transfer learning, which balances performance with lower computational requirements.

Thus, although Vision Transformer (ViT)-based models demonstrate slightly higher accuracy, CNN-based models exhibit greater cost-effectiveness and scalability [45,78-86,90,100,101,107], primarily due to their lower computational demands and reduced operational costs (Figs. 4 and 5).

Scalability is also a key advantage of CNNs, with studies highlighting their adaptability to different hardware configurations and their suitability for real-time applications [76,77,98]. In contrast, ViTs, while powerful, require significantly higher computational resources, making their deployment more costly and less scalable in agricultural settings with infrastructure constraints [96,108,109]. These findings suggest that, while ViTs may be preferable in research environments where accuracy is the primary concern, CNNs remain the more practical choice for real-world applications, where computational efficiency and cost are major constraints.

2) How do data-related factors, such as dataset origin, annotation standards, and size, impact the performance and reliability of AI-based disease detection systems?

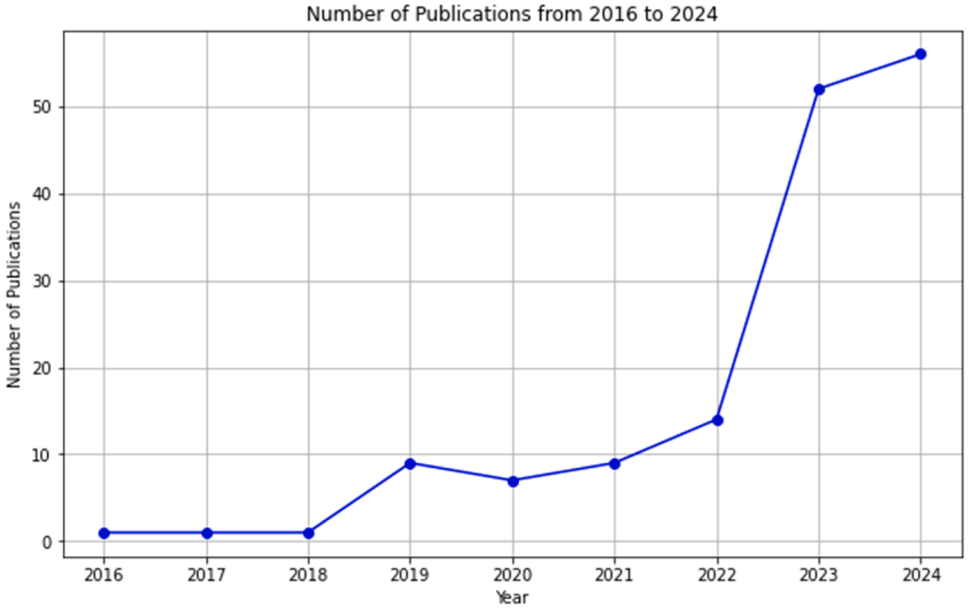


Fig. 2. Increase in the number of publications from 2016 to 2024.

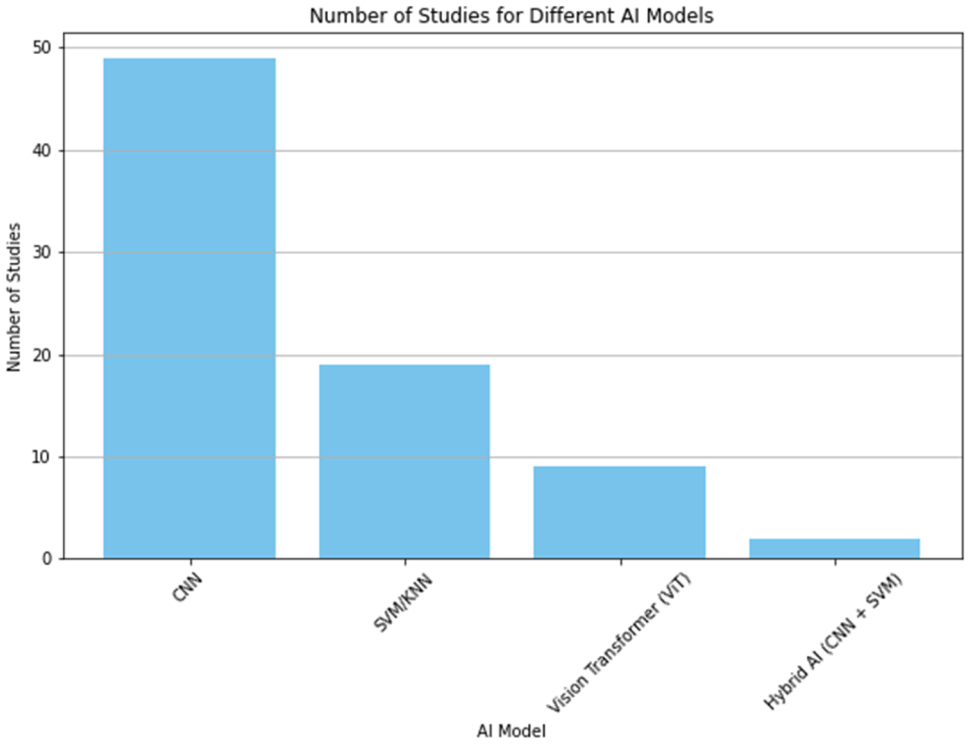


Fig. 3. Number of studies for different AI models.

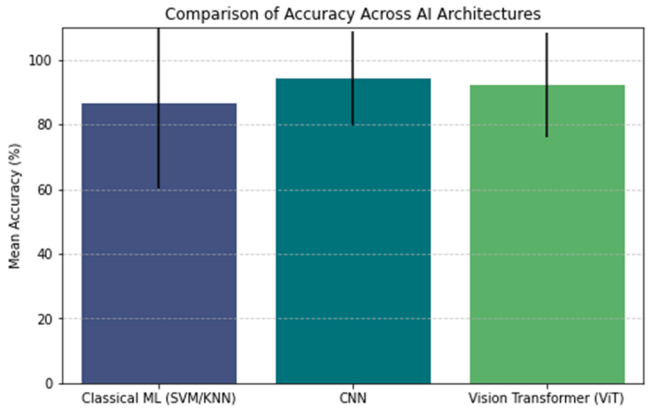


Fig. 4. Comparison of accuracy across AI architectures.

Most of the studies included in this analysis used publicly available image databases from academic repositories or Kaggle. The most widely used public database was PlantVillage [100,110-120, which was employed in 36 studies.

Regarding dataset size, public databases contained an average of approximately 20,311 images, whereas the average size of private datasets was not specified in the included studies. This reliance on public datasets highlights the importance of open-access resources in advancing AI applications in agriculture. In terms of model selection, public datasets are predominantly used in CNN models with transfer learning, leveraging pre-trained architectures for improved performance and efficiency. Conversely, private datasets are more frequently associated with custom AI models, which require greater adaptation to specific agricultural domains and often involve additional preprocessing and augmentation techniques to optimize performance.

Dataset size and quality are critical determinants of performance. However, we should be careful with models trained on large, well-annotated datasets, such as PlantVillage, that demonstrated high accuracy and robustness, because it may rise concerns about overfitting due to controlled laboratory conditions.

The geographic distribution of datasets also revealed a strong concentration in Asian countries, particularly India with 30 studies, China with 15 studies, Pakistan with 5 studies, Bangladesh with 4 studies, and the Philippines, which may introduce a regional bias when attempting to apply these models to other agricultural environments. Beyond Asia, datasets were also sourced from Canada and France, both with 2 studies, though these regions were significantly less represented in the analysis.

RGB images are the most frequently used and represent the most cost-effective type of image acquisition for plant disease detection. This is because RGB cameras are widely available, relatively inexpensive, and do not require specialised imaging equipment. However, some studies have explored the use of high-precision imaging techniques, such as hyperspectral and multispectral imaging, which provide richer spectral information but come at a higher cost due to the need for specialised cameras and sensors.

Even so, studies that utilised high-precision images do not necessarily achieve higher accuracy than those relying on RGB images. The average accuracy for RGB-based models was 96.94 %, whereas models using hyperspectral and multispectral imaging achieved an average accuracy of 95.81 %. High-precision imaging techniques require significantly greater computational resources as they generate large, high-dimensional datasets that necessitate advanced processing techniques and higher memory capacity. Additionally, these methods often require specific environmental conditions for capturing optimal spectral data, which can limit their real-world applicability. This makes RGB images the preferred choice for large-scale, real-world applications where cost-effectiveness, scalability, and ease of deployment are prioritized over maximum accuracy.

Studies that use larger datasets not necessarily tend to achieve higher model accuracy, even when employing data augmentation techniques such as synthetic image generation to artificially expand smaller datasets. This highlights the importance of not only dataset size, but also data quality in enhancing model generalization and robustness.

As shown in Fig. 6, there isn't a significant correlation between

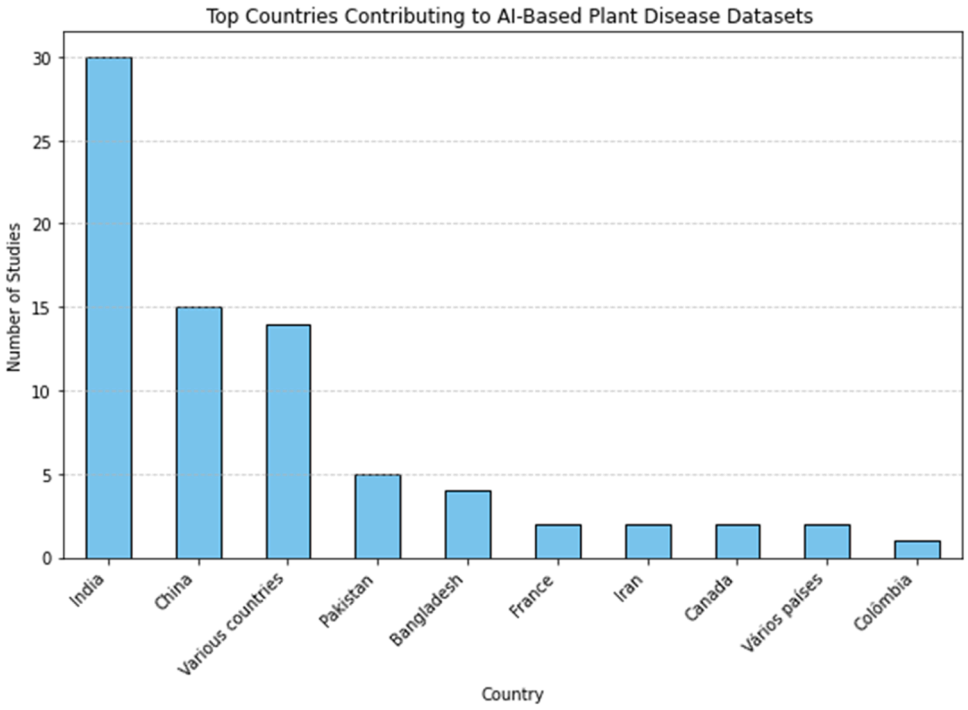


Fig. 5. Top countries contributing to AI-based plant disease datasets.

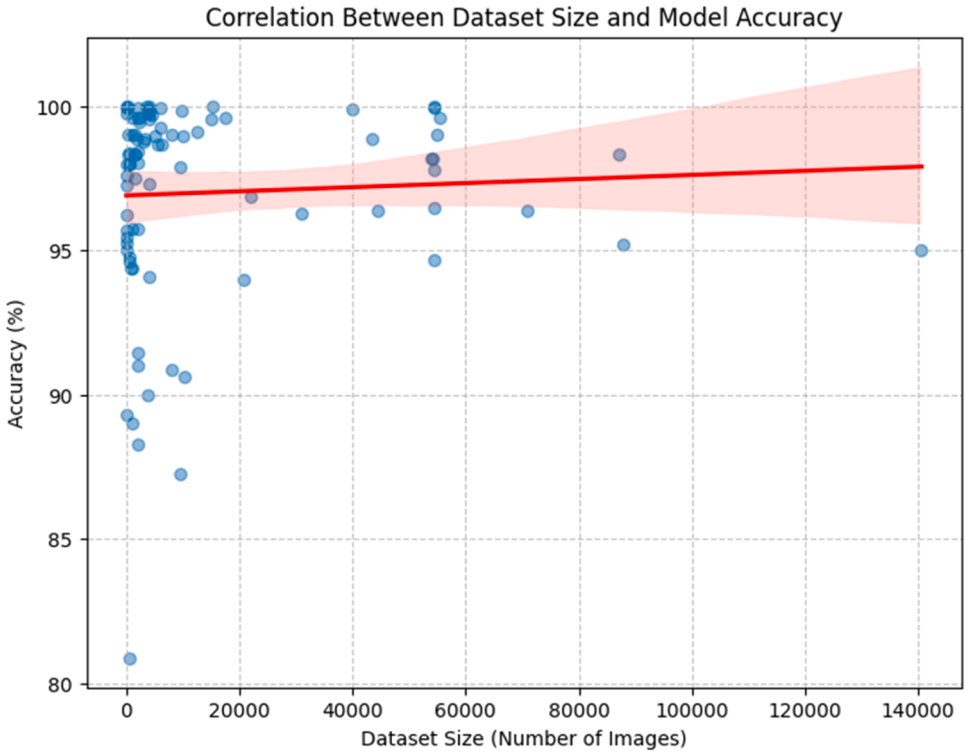


Fig. 6. Correlation between dataset size and model accuracy.

dataset size and model performance, indicating that models trained on larger datasets do not necessarily exhibit higher accuracy. However, this relationship is not strictly linear, as factors such as dataset quality, annotation standards, and feature diversity also play a critical role in determining model effectiveness. In some cases, smaller but well-annotated and diverse datasets can achieve high accuracy.

3) Which types of trees and crops have been most extensively studied in disease detection using artificial intelligence and image analysis, and what factors explain the disparity in research volume among them?

Most of the studies included in this review focused on plants such as tomato, which was included in 31 studies, rice, which was included in 18 studies, grape, which was present in 8 studies, and potato, which was included in 8 studies. A summary is presented in Fig. 7.

This trend can be explained by the social and economic importance

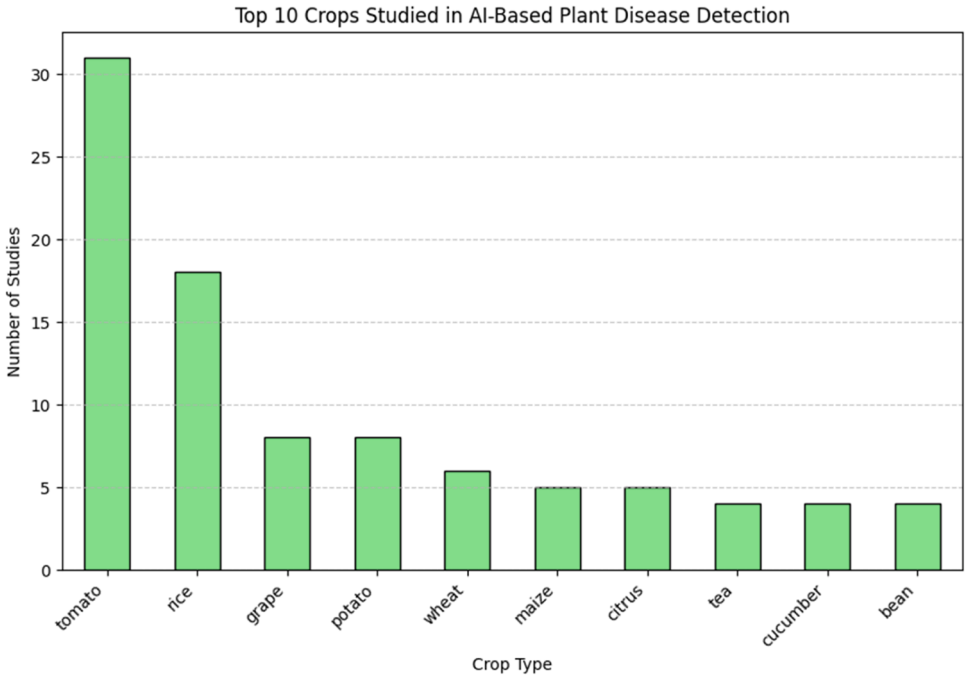


Fig. 7. Top 10 crops in AI-based plant disease detection.

of these staple crops, which play a crucial role in global food security and represent the primary food sources for large portions of the population. Additionally, these crops are widely available in public agricultural databases, particularly PlantVillage and other open-access datasets, which facilitates AI model training and benchmarking. The accessibility of these datasets encourages more research in this domain, as opposed to less commonly studied crops, which often lack well-annotated, publicly available image datasets.

Other crops, such as tea and grapes, are also widely studied due to their economic significance as high-value crops and their impact on global agricultural markets. These crops are particularly sensitive to diseases, making early detection essential for preserving yield quality and preventing substantial economic losses. For example, tea and grapevine diseases can cause severe reductions in production, directly affecting international trade and local economies.

In terms of diseases detected, most studies focused on fungal diseases (97 studies), bacterial infections (65 studies), and viral diseases (20 studies). A total of 16 studies include all three types of diseases: fungal, bacterial, and viral. This indicates a strong research emphasis on fungal plant diseases, among the most prevalent and economically significant crop threats. However, even if more studies focus on fungal infections, the accuracy for fungal disease detection (mean 88.85 % and SD 22.92) is lower than for bacterial (mean 91.39 and SD 18,69) or viral infection (mean 94.16 and SD 10.68). The results are shown in Fig. 8.

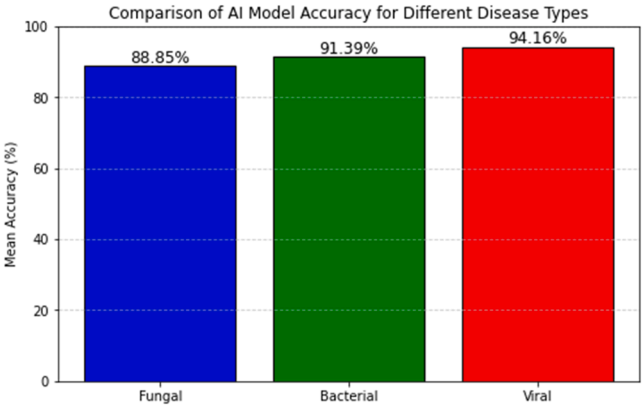


Fig. 8. Comparison of AI model accuracy for difefrent disease types.

Discussion and future directions

Based on the results presented in the previous section, several gaps in the current knowledge have been identified, allowing for the proposal of future research directions and improvements. The first gap concerns geographical bias, as there is a disproportionate concentration of databases originating from the Asian continent compared to other regions. This imbalance may lead to artificial intelligence models with limited generalizability to other areas with different cultural, climatic, and pest/disease pressures. Given this geographic imbalance, there is a clear need for more diverse and representative datasets from underrepresented regions, such as Africa, Europe, and South America. Expanding dataset diversity will improve the robustness, adaptability, and generalisability of AI-based plant disease detection models, ensuring that the methodologies developed can be effectively applied across different climates, crop types, and agricultural conditions.

The analyzed data indicate that the size of databases does not directly correlate with better model performance, particularly in terms of accuracy. Moreover, many studies rely on images from public RGB-format databases, which are not necessarily high-quality and may be insufficient to capture the full complexity of plant diseases. Therefore, future research should focus not only on the number of images or the size of databases but also on image diversity, image quality, and annotation accuracy to enhance the robustness of AI models. Concretely, we recommend the adoption of a standard disease ontology (e.g., PlantVillage taxonomy) and publishing clear labelling protocols, using multi-expert consensus with inter-annotator-agreement metrics (such as Cohen's $\kappa \geq 0.80$) to detect label noise. To ensure image quality, we suggest capturing high-resolution images ($\geq$ 12 MP) under controlled illumination or with portable light tents, from multiple viewing angles and growth stages. We also recommend recording contextual metadata such as location, soil parameters, weather, and phenological stage to aid domain adaptation. These measures can improve annotation reliability

and image fidelity, leading to more generalisable AI models for plant-disease detection.

The models that achieved the best performance were those based on deep learning, such as CNNs and Vision Transformers (ViT), which generally outperform traditional machine learning models like SVM and KNN. However, deep learning models are also the most computationally expensive and challenging to implement, which can limit their scalability. Additionally, a notable underrepresentation of hybrid models integrating deep learning and traditional machine learning was observed. In this regard, future research could explore hybrid approaches that leverage the advantages of both methods, balancing lower computational complexity with high accuracy.

Cost-effectiveness analysis is not reported in most studies, and data on this aspect remain scarce. However, CNN-based models demonstrate relative robustness and can operate with RGB images, without requiring high-resolution inputs such as multispectral or hyperspectral images. Therefore, future development should focus on low-cost models and infrastructures to facilitate implementation, even in resource-constrained markets. Additionally, these techniques and models should be designed for ease of use, preferably through cloud computing, to minimize investment requirements (e.g., the need for high-performance computers and GPUs). The adoption of open-source software further enhances collaboration, accessibility, and research advancement in the field.

The results of this review demonstrated that deep learning models exhibit high robustness in detecting crop diseases. However, it is important to highlight that the investigated models were highly specific and not tested in diverse environments, such as those with a wide variety of species, different climatic conditions, or pest outbreaks. The adaptation of pre-existing models to new environments through transfer learning should continue to be explored, and greater diversity should be incorporated into testing. This approach can enhance the adaptability of previously trained models, thereby reducing the financial and computational costs associated with training new models from scratch.

Disparities were also observed regarding the distribution of studied crops, with a predominance of research focused on rice, potatoes, and tomatoes. While this trend may be partially explained by their economic importance and role in food security, expanding research to other crops is essential to ensure a more comprehensive and sustainable approach to agricultural resilience. Additionally, fungal diseases received significantly more attention in the analyzed studies compared to bacterial or viral infections. Future research should aim to balance this focus and prioritize the study of less-known diseases that may pose critical economic and food security risks, particularly in vulnerable and underprotected regions.

Finally, it is crucial to incorporate and discuss the risks associated with technological solutions, particularly concerning cybersecurity and the vulnerability to cyberattacks, whether perpetrated by hackers or even by governments in times of conflict. These technologies are especially susceptible to risks such as data manipulation, information theft, and deliberate system shutdowns, among others. Therefore, future research should also focus on digital security, emphasizing the development of secure methods for data transmission and storage, ensuring data integrity, and protecting models against external threats.

Conclusion

Currently, deep-learning models, particularly CNNs and ViTs, deliver the most accurate results for detecting crop diseases. Further progress depends on creating diverse, well-labelled datasets and building lightweight systems that work in low-resource farm settings. Closing geographic data gaps and adopting common annotation standards are key to turning research success into real-world impact. Future studies should prioritize the development of hybrid AI models that combine the efficiency of classical machine learning with the accuracy of deep learning, aiming to balance performance with computational demands. Greater attention must also be given to the creation of open, multi-regional datasets that capture a wide diversity of crops, disease types, and environmental conditions to enhance model generalizability. Another important direction is the design of cost-effective and scalable solutions, that can support smallholder farmers in resource-constrained contexts. Research should expand into cybersecurity safeguards for agricultural AI pipelines, ensuring resilience against adversarial attacks and data manipulation. By addressing these dimensions, future work will accelerate the transition of AI-based disease detection from experimental studies to practical, secure, and widely accessible agricultural tools.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

If this manuscript involves research on animals or humans, it is imperative to disclose all approval details.

CRediT authorship contribution statement

Munir Majdalawieh: Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Carla Martins:** Writing – original draft, Visualization, Formal analysis. **Mohammed Radi:** Supervision, Methodology, Investigation, Conceptualization. **Maher Alaraj:** Visualization, Validation, Conceptualization. **Shafaq Khan:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2025.101491.

Data availability

No data was used for the research described in the article.

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